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( 10 M
[@01]

## QUESTION ONE

Choose the most correct answer

I. Which of the following is a characteristic of digital signals?
   - A. Continuous variation of signal amplitude
   - B. Represented using discrete levels or states
   - C. More susceptible to compared to analog signals
   - D. Difficult to store and p using computers

II. What is the main advantage of digital electronics over analog electronics?
   - A. Lower power consumption in all cases
   - B. Ability to handle high-frequency signals easily
   - C. Greater resistance to nois signal degradation
   - D. Easier design of high-p amplifiers

III. Which component is most commonly associated with digital electronics?
   - A. Operational amplifiers
   - B. Resistors
   - C. Microprocessors
   - D. Transformers

IV. In digital systems, a binary "1" typically represents which of the following?
   - A. A high voltage level
   - B. A low voltage level
   - C. A continuously varying signa
   - D. An undefined stat

V. Which of the following statements is true about analog signals?
   - A. They have discrete levels or steps for signal representation.
   - B. They are more suitable for long-distance transmission without degradation.
   - C. They can take any value with given range.
   - D. They are always represented binary form.

VI. In the 2's complement form, the binary number 01101101 is equal to the decimal numb
   - A. -19
   - B. +109
   - C. +91
   - D. -109

10010011

VII. The binary number for $F7C9_{16}$ is
   - A. $1111011110101001_2$
   - B. $1111011111001001_2$
   - C. $1110111110101001_2$
   - D. $1111011010101001_2$

9-10
12-1

VIII. The BCD number for decimal 473 is
   - A. 111011010
   - B. 110001110011
   - C. 010001110011
   - D. 010011110011

7-0
F-15-1

IX. The Boolean expression X = (A + B)(C + D) represents
   - A. Two ORs ANDed together
   - B. Two ANDs ORed together
   - C. A 4-input AND gate
   - D. A 4-input OR gate

X. An example of a standard SOP expression is
   - A. $\bar{A}B + A\bar{B}C + AB\bar{D}$
   - B. $A\bar{B}C + A\bar{C}D$
   - C. $A\bar{B} + \bar{A}B + AB$
   - D. $A\bar{B}C\bar{D} + \bar{A}B + A$

## QUESTION TWO

**(10 Marks)**

i. Express -87 decimal number as an 8-bit number in the 1's complement form **[03 Marks]**



10011
11000

REG No: ______________________________

ii. Convert 56 and 227 decimal numbers to binary and add using the 2's complement form **[03 Marks]**

iii. Develop a circuit with inputs *A*,*B*,*C*,*D* where the output is **HIGH** if *A* is **HIGH**, or B,C, and D are all **HIGH**. **[04 Marks]**

09/10

**(20 Marks)**

## QUESTION THREE

i. Simplify the following Boolean expression using Boolean laws and rules **[05 Marks]**

$$[A\overline{B}(C + BD) + \overline{A}\,\overline{B}]C$$

ii. Use a Karnaugh map to find the minimum SOP form for the expression below **[05 Marks]**

$$A + B\overline{C} + CD$$

iii. Use NAND gates to implement the following logic expression **[05 Marks]**

$$\overline{A}B + CD + \overline{(A + B)}(ACD + \overline{BE})$$

iv. Use NOR gates to implement the following logic expression **[05 Marks]**

$$AB\overline{C}\overline{D} + D\overline{E}F + A\overline{F}$$

09/10

**QUESTION ONE - ANSWER**

| I | II | III | IV | V | VI | VII | VIII | IX | X |
|---|---|---|---|---|---|---|---|---|---|
| B | C | ~~A~~ | A | C | B | B | C | A | C |

09/10

DEPARTMENT OF ELECTRONICS AND TELECOM...

SEMESTER I: TEST II - 2024/2025

ET 8301: DIGITAL ELECTRONICS

BEng 3rd YEAR - TELECOMMUNICATION SYSTEM ENGINEERING

TIME: 60 Minutes

DATE: 26th January, 2025

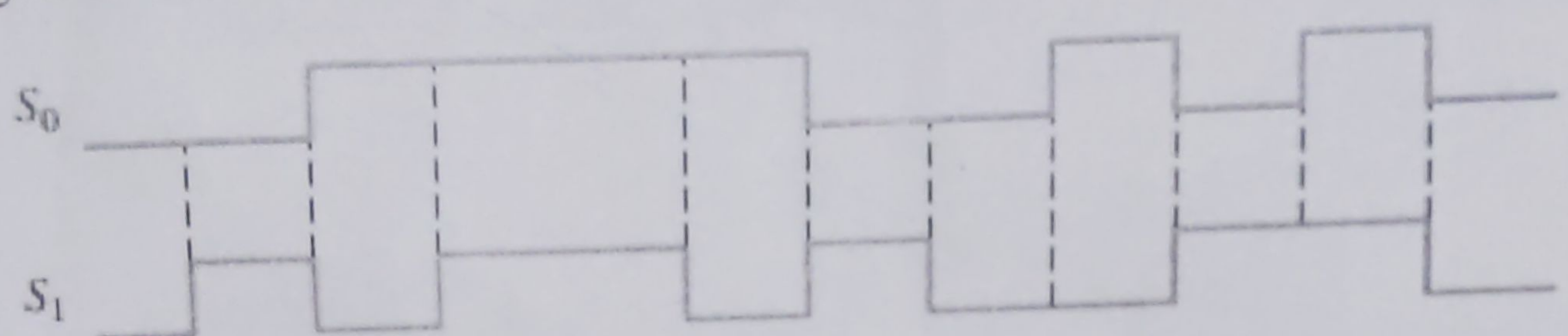
INSTRUCTIONS

1. Answer **ALL** questions.

## QUESTION ONE

(02.5 Marks)

If the data-select inputs to the 4 to 1 multiplexer are sequenced as shown by the waveforms in Figure below, determine the output waveform with the data inputs $D_0 = 1$, $D_1 = 0$, $D_2 = 0$, $D_3 = 1$.




## QUESTION TWO

(02.5 Marks)

Convert the following SOP expression to an equivalent POS expression:

$$\overline{A}\,\overline{B}\,\overline{C} + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}C + ABC$$

REG No: ____________________

## QUESTION THREE

**(05 Marks)**

Determine the Q and Q output waveforms of the flip-flop in Figure A for the D and CLK inputs in Figure B, Assume that the positive edge-triggered flip-flop is initially RESET

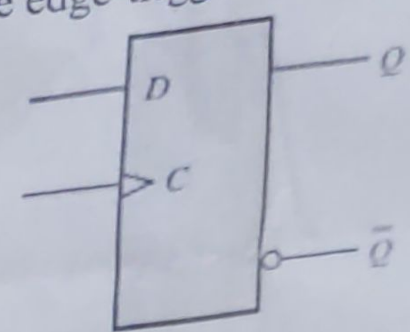



Figure A

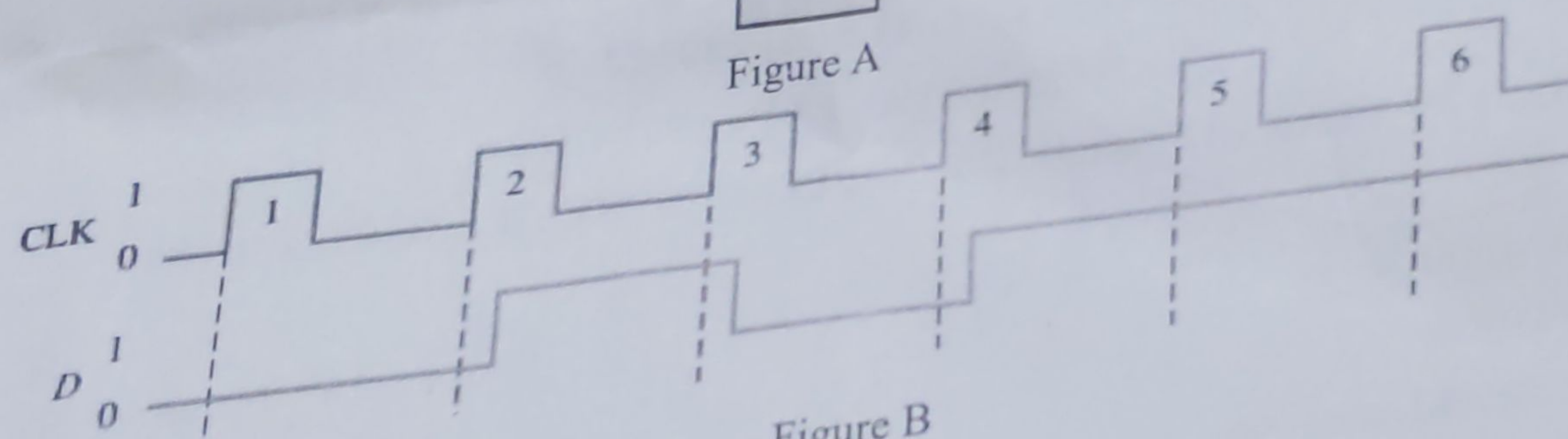



Figure B

## QUESTION FOUR

**(05 Marks)**

Design a counter with the binary count sequence 1,2,5,7. Use D flip-flops.